WORKED SOLUTION

STEP II 2015, Question 12

Probability and Statistics • Coin toss

STEP II 2015, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q12

The key idea

Every sequence here has length three, so whether a toss ends the game depends only on that toss and the two before it. The natural thing to track is therefore the *last two tosses*: there are just four possible states (HH, HT, TH, TT), and from each one the next toss either ends the game or moves us to another state. Some states settle the winner at once; for the others we set up a pair of simple equations by conditioning on the next toss.

Solution

Throughout, tosses are independent, and each is H or T with probability $\frac{1}{2}$. No sequence can be completed in the first two tosses, so the game always reaches one of the four two-toss states, each with probability $\frac{1}{4}$. Note too that the game ends with probability 1: from any position, the next three tosses complete a given sequence with probability $\frac{1}{8}$, so the chance of the game lasting more than $3n$ further tosses is at most $(\frac{7}{8})^n \to 0$.

Part (i)

Here only HHT (for A) and THH (for B) count.

If the first two tosses are HH: the coin keeps showing H for a while, and the first T to appear completes HHT, since it is preceded by at least two Hs. Before that T no tail has appeared, so THH cannot have occurred. So A wins.

If a T appears among the first two tosses: consider the first time two consecutive heads appear after that T. The toss just before this HH cannot be H (otherwise there would be an earlier HH after the T, or the HH would be the opening pair, which we've ruled out), so it is T and THH has been completed. HHT can't have occurred earlier, because it needs an HH, and this is the first one. So B wins.

Hence A wins exactly when the first two tosses are both heads, and

$$\mathrm{P}(A \text{ wins}) = \tfrac{1}{2} \times \tfrac{1}{2} = \tfrac{1}{4},$$

as required.

Part (ii)

Now all four sequences HHT, THH, TTH, HTT are live. Condition on the first two tosses.

- **HH.** As in part (i), the tosses continue H, H, … until the first T, which completes HHT. None of the other sequences contains HH followed by H, and no tail has appeared to start THH, TTH or HTT. So A wins.
- **TT.** By the same argument with the roles of H and T swapped, the first H completes TTH, so C wins.
- **HT.** If the next toss is T we have HTT and D wins. If it is H, the last two tosses are TH (and HTH completes nothing).
- **TH.** If the next toss is H we have THH and B wins. If it is T, the last two tosses are HT (and THT completes nothing).

So once the game is in state HT or TH it bounces between these two until D or B wins. Let d_1 and d_2 be the probabilities that D wins starting from HT and from TH respectively. Conditioning on the next toss,

$$d_1 = \tfrac{1}{2} + \tfrac{1}{2}d_2, \qquad d_2 = \tfrac{1}{2}d_1.$$

Substituting, $d_1 = \frac{1}{2} + \frac{1}{4}d_1$, so $d_1 = \frac{2}{3}$ and $d_2 = \frac{1}{3}$. From these states only B or D can win, so B wins with probability $\frac{1}{3}$ from HT and $\frac{2}{3}$ from TH. Therefore

$$\mathrm{P}(A) = \tfrac{1}{4}, \qquad \mathrm{P}(C) = \tfrac{1}{4},$$
$$\mathrm{P}(D) = \tfrac{1}{4}\cdot\tfrac{2}{3} + \tfrac{1}{4}\cdot\tfrac{1}{3} = \tfrac{1}{4}, \qquad \mathrm{P}(B) = \tfrac{1}{4}\cdot\tfrac{1}{3} + \tfrac{1}{4}\cdot\tfrac{2}{3} = \tfrac{1}{4}.$$

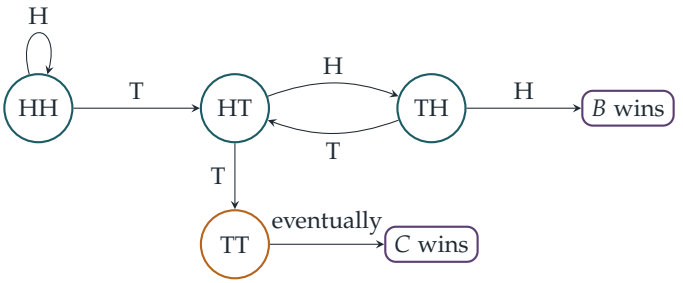
Each player wins with probability $\frac{1}{4}$.

Part (iii)

Now only THH (for B) and TTH (for C) count.

First two tosses TT. The tosses then run T, T, … until the first H, which completes TTH. THH needs a TH followed by H, and no H has appeared yet, so B can't have won first. Hence C wins with probability 1.

For the other starts it helps to see how the state (the last two tosses) moves, given that the game hasn't ended:



(From TT the next T keeps us in TT and the next H completes TTH, which is why we can treat TT as a win for C.)

With p, q, r the probabilities that C wins from HT, TH, HH:

From HT. A T gives TT, from which C wins with probability 1; an H gives HTH, which completes neither sequence and leaves us in state TH. So

$$p = \tfrac{1}{2} \cdot 1 + \tfrac{1}{2} \cdot q = \tfrac{1}{2} + \tfrac{1}{2}q,$$

as required.

From TH. An H completes THH, so B wins; a T gives THT, completing nothing, and we are in state HT. So $q = \tfrac{1}{2} \cdot 0 + \tfrac{1}{2}p = \tfrac{1}{2}p$.

Ask yourself Why does the past history not matter once we know the last two tosses?

Both target sequences have length three, so whether the next toss finishes the game depends only on that toss and the previous two. Earlier tosses are irrelevant, and future tosses are independent of the past. So the chance that C wins from a given state is the same however we got there, which is what lets q reappear on the right-hand side of the equation for p.

From HH. An H leaves us in HH; a T moves us to HT. So $r = \tfrac{1}{2}r + \tfrac{1}{2}p$, giving $r = p$.

Solving: $p = \tfrac{1}{2} + \tfrac{1}{4}p$, so

$$p = \tfrac{2}{3}, \qquad q = \tfrac{1}{3}, \qquad r = \tfrac{2}{3}.$$

Finally, conditioning on the first two tosses, each with probability $\tfrac{1}{4}$,

$$\mathrm{P}(C \text{ wins}) = \tfrac{1}{4}(1 + p + q + r) = \tfrac{1}{4}\left(1 + \tfrac{2}{3} + \tfrac{1}{3} + \tfrac{2}{3}\right) = \tfrac{1}{4} \cdot \tfrac{8}{3} = \tfrac{2}{3}.$$

Checks

- In part (ii) the four probabilities add to 1, and the answer is symmetric: swapping H and T exchanges $A \leftrightarrow C$ and $B \leftrightarrow D$, so $\mathrm{P}(A) = \mathrm{P}(C)$ and $\mathrm{P}(B) = \mathrm{P}(D)$ had to happen.
- In part (iii), $q = \tfrac{1}{3} < p$ makes sense: from TH, B is one toss away from winning.

- A computer simulation of 200,000 games of each type gave $\mathrm{P}(A) \approx 0.250$ in part (i), all four players close to 0.25 in part (ii), and $\mathrm{P}(C) \approx 0.667$ in part (iii), matching exact calculations along the Markov chain.

Common slips

- Assuming that two sequences of the same length must be equally likely to appear first. Each of HHT and THH has expected waiting time 8 tosses on its own, yet THH beats HHT three times out of four.
- Forgetting that a non-winning toss can still leave you in a useful state: from HT, the toss H doesn't just "fail", it takes you to TH.
- Writing $p = \frac{1}{2} + \frac{1}{2}q$ without explaining why HTH leaves you in the same position as starting from TH.

Beyond STEP

The method here is a first step into *Markov chains*: a process whose future depends only on its current state, analysed by solving linear equations for absorption probabilities. This particular game is a version of Penney's game, in which, for any sequence of three tosses your opponent chooses, you can always pick one that beats it with probability at least $\frac{2}{3}$, so "which sequence comes first" is not a transitive relation. Grimmett and Welsh, *Probability: An Introduction*, treats Markov chains and absorption probabilities at just this level.

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